PV101 Day 7 Master Print Packet

Compiled packet

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Contents follow in instructional order.

PV101 Day 7 Daily Run Sheet

Instructor-facing plan for Rich or a substitute | Wednesday, July 15, 2026

Day_07_PV101_Load_Assessment_and_System_Sizing_2026-07-15

Topic: Load Assessment and System Sizing

Class setup: 6 students | 10:00 AM-3:00 PM | Calm, practical, evidence-first instruction

Teaching frame

- Main message: loads drive system sizing, and units make the reasoning inspectable.
- Day 7 is guided practice, not a project, proposal, capstone, or presentation.
- Use only the documented training values on the worksheet; do not invent equipment or product specifications.
- Safety remains part of design: protection, disconnects, labels, and verification points must appear on the one-line diagram.

Substitute quick start

- [ ] Read the full schedule before students arrive.
- [ ] Locate emergency disconnects, first-aid supplies, calculator policy, and secure test storage.
- [ ] Print student materials and keep answer keys/instructor notes separate.
- [ ] Write the day goal, schedule, and stop-work rule on the board.
- [ ] If uncertain about equipment or assessment handling, use the reduced-scope path and contact program leadership.

Learning targets

- Calculate daily energy use in watt-hours from a documented loads list.
- Use provided assumptions to calculate basic battery Wh/Ah and array watts.
- Choose an inverter rating from documented simultaneous AC loads and headroom.
- Draw and label a one-line system path with protection, disconnects, and verification points.
- Check units, assumptions, and reasonableness before instructor signoff.

Materials

- Chapters 11-12 quiz and key
- Day 7 slide deck and whiteboard outline
- Student worksheet and guided-practice guide
- Calculators, pencils, rulers, scrap paper
- Documented RV training loads and provided assumptions
- Evidence checklist and reflection sheet

Daily schedule

Time	Block	Instructor notes
10:00-10:10	Arrival, attendance, check-in	Students get notebooks and Chapters 11-12 quiz. Instructor displays: Measure. Verify. Document. Then proceed.
10:10-10:20	Safety reset + day goal	Review stop-work authority and explain that today is guided sizing practice, not a project or product proposal.
10:20-10:35	Morning quiz	Individual Chapters 11-12 reading check. Ten questions; quiet work.
10:35-10:55	Quiz correction	Correct together in a different color. Review current-source verification, AHJ/permitting roles, units, and safest-answer thinking.
10:55-11:20	Whiteboard review	Connect Chapters 1-12: loads drive sizing; safety and verification constrain every decision.
11:20-11:30	Break	Reset attention. Calculators, worksheets, and pencils staged after break.
11:30-12:00	Slides + worked example	Model watts x hours, total Wh/day, battery Wh/Ah, array watts, and inverter headroom using the provided training values.
12:00-12:30	Guided example check	Students complete one example with instructor pauses after each calculation. Check units before accepting answers.
12:30-1:00	Lunch	Worksheets remain on tables. No kit wiring or energized work.
1:00-1:15	Practice introduction	Explain the RV training load worksheet, required assumptions, evidence boxes, and instructor checkpoints.

1:15-1:50	Guided worksheet: loads + battery	Students calculate daily Wh and basic storage. Instructor checks totals before array sizing.
1:50-2:15	Guided worksheet: array + inverter	Students calculate array watts and identify concurrent AC load plus headroom.
2:15-2:30	Diagram + protection points	Students draw a one-line system path and mark conceptual protection, disconnects, labels, and meter points.
2:30-2:45	Verification + revision	Peer checks math, units, assumptions, and diagram. Instructor signs only after corrections.
2:45-2:55	Reflection + evidence upload	Complete reflection and upload/submit the worksheet as directed.
2:55-3:00	Cleanup + dismissal	Calculators returned, papers collected, tables clear, next-day final reminder given.

Safety and instructional stop conditions

- A student uses an assumption or specification not provided in the worksheet.
- Units are missing or mixed without conversion.
- A sizing answer is copied without work or cannot be explained.
- The diagram omits protection, disconnects, or return paths.
- Discussion drifts into physical wiring; this day is paper/calculator practice.

Reduced-scope backup

Day 7: if calculators or printing fail, complete one class example on the whiteboard and have students show units and reasonableness on notebook paper. Day 8: if final-test printing fails, run structured review and reflection; do not improvise a different final from memory. Reschedule the secured assessment.

Early finishers

Day 7: check every unit, redo one calculation independently, and write one sentence explaining each sizing result. Day 8: quietly review answers, then begin the reflection only after turning in the final.

Instructor notes after class

What landed:

_____

_____

What confused students:

_____

_____

Equipment/material issues:

_____

_____

Carry-forward:

_____

_____

PV101 Day 7 Morning Quiz

Student copy | 10-15 minutes

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Name: _____ Date: _____

Answer briefly. Show work and units where needed.

1. What does AHJ stand for in a permitting context?

_____

_____

2. Why should the AHJ and utility requirements be checked before an installation begins?

_____

_____

3. Why must an incentive be verified with a current authoritative source?

_____

_____

4. Name two useful authoritative resources for planning permitted PV work.

_____

_____

5. What is the purpose of a permit and inspection process?

_____

_____

6. What should you do first when a practice question includes units?

_____

_____

7. When two answers seem possible, what type of answer takes priority in PV work?

_____

_____

8. Why should a calculated answer be checked for reasonableness?

_____

_____

9. Why is reviewing a missed practice question useful?

_____

_____

10. State the course rule that replaces guessing.

_____

_____

PV101 Day 7 Morning Quiz Answer Key

Instructor copy

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1. What does AHJ stand for in a permitting context?
Authority Having Jurisdiction.

2. Why should the AHJ and utility requirements be checked before an installation begins?
They define the current local approval, documentation, inspection, and interconnection requirements that apply to the work.

3. Why must an incentive be verified with a current authoritative source?
Programs, eligibility rules, amounts, and deadlines can change.

4. Name two useful authoritative resources for planning permitted PV work.
Accept two: the local AHJ/building department, the serving utility, current government program material, adopted code or local amendments, or manufacturer instructions.

5. What is the purpose of a permit and inspection process?
To document that the proposed and completed work meets the applicable local safety and approval requirements.

6. What should you do first when a practice question includes units?
Identify and keep track of the units before calculating or selecting an answer.

7. When two answers seem possible, what type of answer takes priority in PV work?
The answer that preserves safety, verification, and documented procedure.

8. Why should a calculated answer be checked for reasonableness?
A correct-looking calculation can still contain a wrong unit, decimal, or assumption.

9. Why is reviewing a missed practice question useful?
It identifies the concept, unit, calculation, or safety reasoning that needs correction.

10. State the course rule that replaces guessing.
Measure, verify, document, then proceed; equivalent wording accepted.

PV101 Day 7 Whiteboard Outline

Instructor-facing

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Board before students enter

- PV101 Day 7 | Chapters 11-12 | Loads drive sizing
- Chapter 11: verify incentives, resources, permitting, and the AHJ.
- Chapter 12: practice, review misses, and show the work.
- W = V x A | Wh = W x h
- Measure. Verify. Document. Then proceed.

Worked-example sequence

- 1. List loads and hours.
- 2. Calculate Wh/day.
- 3. Apply autonomy and usable fraction.
- 4. Convert Wh to Ah at system voltage.
- 5. Divide daily Wh by peak sun hours and apply the provided loss multiplier.
- 6. Choose inverter headroom from concurrent AC load.
- 7. Draw protection, disconnects, labels, and test points.

Questions to ask

- What unit should this answer have?
- Which value came from the worksheet and which did you calculate?
- Is the result reasonable?
- What would change if the hours or loads changed?

PV101 Day 7 Student Worksheet

Load assessment and basic system sizing | Guided practice

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Name: _____ Date: _____

Operating rules

- Use only the values provided.
- Show formulas, numbers, and units.
- Circle assumptions.
- Check reasonableness before asking for signoff.
- This is guided practice—not a product design or proposal.

Provided training assumptions

Item	Training value
System voltage	12 V
Usable battery fraction / maximum depth of discharge	80% for this exercise
Autonomy for this exercise	1 day
Peak sun hours for this exercise	4 h/day
Loss multiplier for this exercise	1.25
Inverter headroom for this exercise	25% above concurrent AC load

Part A: RV training loads

Load	Power	Hours/day	Calculation	Wh/day
3 interior LED lights	8 W each	4 h		
Water pump	60 W	0.5 h		
Vent fan	36 W	3 h		

Phone/USB charging	24 W	2 h
Laptop charger (AC)	65 W	2 h

Total daily energy: _____ Wh/day

Part B: Storage

Required stored Wh = daily Wh x autonomy = _____

_____

Battery nameplate Wh = required stored Wh / usable fraction = _____

_____

Battery Ah at 12 V = battery nameplate Wh / 12 V = _____

_____

Reasonableness sentence: _____

_____

Part C: Array and inverter

Raw array watts = daily Wh / peak sun hours = _____ W

_____

Adjusted array watts = raw array watts x 1.25 = _____ W

_____

Concurrent AC load used for inverter sizing: _____ W

_____

Minimum inverter rating with 25% headroom: _____ W

_____

What load or assumption most affects the result? _____

_____

Part D: One-line diagram

Draw: PV -> controller -> battery -> inverter/DC loads. Show return paths, protection, disconnects, labels, and meter points.

_____

_____

_____

_____

Verification and signoff

Check	Peer	Instructor
Loads and hours copied correctly		
Every calculation shows units		
Assumptions match provided values		
Answers are reasonable and rounded clearly		
Diagram includes both directions/return path		
Protection, disconnects, labels, and test points shown		

PV101 Day 7 Student Worksheet Answer Key

Instructor copy

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RV training loads

Load	Calculation	Wh/day
3 interior LED lights	3 x 8 W x 4 h	96
Water pump	60 W x 0.5 h	30
Vent fan	36 W x 3 h	108
Phone/USB charging	24 W x 2 h	48
Laptop charger (AC)	65 W x 2 h	130
Total	96 + 30 + 108 + 48 + 130	412 Wh/day

Storage

- Required stored energy: 412 Wh/day x 1 day = 412 Wh.
- Battery nameplate energy: 412 Wh / 0.80 = 515 Wh.
- Basic capacity at 12 V: 515 Wh / 12 V = 42.9 Ah.
- Accept clear rounding. Students must state the 1-day autonomy and 80% usable-fraction / maximum-depth-of-discharge assumptions.

Array and inverter

- Raw array watts: 412 Wh / 4 peak sun hours = 103 W.
- Adjusted exercise result: 103 W x 1.25 = 128.75 W, or about 129 W.
- Concurrent AC load: 65 W for the laptop charger; the other listed loads are treated as DC in this exercise.
- Minimum inverter exercise result: 65 W x 1.25 = 81.25 W.
- Do not turn the result into a product recommendation. Actual selection needs real specifications, surge data, and documented design inputs.

One-line diagram acceptance

Accept a clearly labeled path such as PV -> charge controller -> battery -> protected DC load/inverter branches. Require complete return paths, source-side protection, disconnects, polarity/labels, and logical meter points. Conceptual variants are acceptable when electrically coherent and safely explained.

PV101 Day 7 Guided-Practice Guide

Student-facing

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Name: _____ Date: _____

Purpose

Guide students through a bounded sizing exercise using only the provided RV training loads and assumptions.

Tools and materials

- Student worksheet
- Calculator and pencil
- Ruler
- Provided training assumptions
- No physical PV equipment required

Procedure

- Complete the loads table.
- Checkpoint 1: instructor verifies daily Wh.
- Calculate storage using the provided autonomy and 80% usable-fraction / maximum-depth-of-discharge assumption.
- Checkpoint 2: instructor verifies units and arithmetic.
- Calculate raw and adjusted array watts.
- Select inverter headroom from concurrent AC load.
- Draw and peer-check the one-line diagram.
- Revise and obtain instructor signoff.

Common mistakes

- Adding watts without multiplying by hours
- Confusing W with Wh
- Dividing by the wrong voltage
- Omitting the usable-fraction / depth-of-discharge adjustment
- Sizing inverter from daily energy instead of simultaneous power
- Inventing equipment values not supplied

Cleanup

- Return calculators and rulers.
- Submit worksheet and reflection.
- Clear tables and recycle scratch paper as directed.

PV101 Day 7 Reflection Questions

Student copy

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Name: _____ Date: _____

1. What sizing step makes the most sense to you now?

_____

_____

2. Where are you most likely to make a unit or assumption error?

_____

_____

3. How does verification change a design calculation?

_____

_____

4. What did Days 1-6 teach that helped today?

_____

_____

5. What would you check before trusting someone else's sizing worksheet?

_____

_____

PV101 Day 7 Instructor Notes

Instructor-facing

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Before class

- Print and separate student/instructor copies.
- Verify the 10:00-3:00 schedule and room setup.
- Keep answer keys controlled.

During class

- Use missed quiz items as teaching feedback.
- Require units and evidence.
- Protect student dignity; correct privately and calmly.

After class

- Record what landed and what needs follow-up.
- Secure graded work.
- Complete the evidence checklist and Daily Summary.

PV101 Day 7 Evidence Checklist

Instructor copy

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- [ ] Chapters 11-12 quiz completed
- [ ] Quiz corrections completed
- [ ] Daily Wh total checked
- [ ] Battery Wh/Ah calculation checked
- [ ] Array and inverter calculations checked
- [ ] One-line diagram checked
- [ ] Peer verification completed
- [ ] Instructor signoff completed
- [ ] Reflection submitted
- [ ] Classroom evidence submitted

Instructor initials: _____ Notes: _____